Lab 09 – ListBox, ComboBox, DatePicker

Title: Enhanced Input and Filtering with Specialized Controls

Objectives

This lab introduced more specialized WPF controls that handle specific data types and interactions. I needed to use a ComboBox for selecting from a predefined list of departments, a DatePicker for selecting dates with a calendar interface, and a ListBox for multiple-selection filtering by job function. These controls significantly improve the user experience compared to simple text inputs.

Implementation

MainWindow.xaml (Filter Section)

```
<!-- Filter Controls Section -->
<Border Grid.Row="0" Background="{StaticResource LightBrush}" Padding="10" Margin="0,0,0,10" CornerRadius="5">
  <Grid>
    <Grid.ColumnDefinitions>
      <ColumnDefinition Width="*"/>
      <ColumnDefinition Width="*"/>
    </Grid.ColumnDefinitions>

    <StackPanel Grid.Column="0">
      <TextBlock Text="Filtrare angajați" FontWeight="Bold" FontSize="14" Margin="0,0,0,10"/>

      <!-- Tip Salariu RadioButtons -->
```

```
        <GroupBox Header="Tip Salariu" Margin="0,0,0,10">
            <StackPanel Orientation="Horizontal">
                <RadioButton x:Name="rbToate" Content="Toate" IsChecked="True"
GroupName="TipFiltru" Margin="5" Checked="RadioButton_Checked"/>
                <RadioButton x:Name="rbLunar" Content="Lunar" GroupName="TipFiltru"
Margin="5" Checked="RadioButton_Checked"/>
                <RadioButton x:Name="rbOrar" Content="Orar" GroupName="TipFiltru" Margin="5"
Checked="RadioButton_Checked"/>
            </StackPanel>
        </GroupBox>

        <!-- Bonus CheckBox -->
        <CheckBox x:Name="chkBonus" Content="Doar cu bonusuri" Margin="5"
Checked="CheckBox_Changed" Unchecked="CheckBox_Changed"/>

        <!-- ListBox for Function Filter -->
        <GroupBox Header="Funcție" Margin="0,10,0,0">
            <ListBox x:Name="lstFunctii" SelectionMode="Multiple" Height="100"
                     SelectionChanged="LstFunctii_SelectionChanged">
                <ListBoxItem>Programator</ListBoxItem>
                <ListBoxItem>Manager</ListBoxItem>
                <ListBoxItem>Analist</ListBoxItem>
                <ListBoxItem>Designer</ListBoxItem>
                <ListBoxItem>Contabil</ListBoxItem>
                <ListBoxItem>Resurse Umane</ListBoxItem>
            </ListBox>
        </GroupBox>

        <Button Content="Aplică filtre" x:Name="btnFiltreaza" Click="BtnFiltreaza_Click"
                Background="{StaticResource AccentBrush}" Foreground="White"
                Padding="15,5" Margin="0,10" Width="120" HorizontalAlignment="Left"/>
    </StackPanel>
```

```
    <StackPanel Grid.Column="1" Margin="20,0,0,0">
      <TextBlock Text="Căutare rapidă" FontWeight="Bold" FontSize="14" Margin="0,0,0,10"/>

      <Label Content="Caută după nume:" Margin="0"/>
      <TextBox x:Name="txtCautareNume" TextChanged="TxtCautare_TextChanged"
Margin="0,0,0,10"/>

      <Label Content="Departament:" Margin="0"/>
      <ComboBox x:Name="cmbDepartamentFiltru" FontSize="14" Padding="5"
Margin="0,0,0,10"
                SelectionChanged="CmbDepartament_SelectionChanged">
        <ComboBoxItem>Toate</ComboBoxItem>
        <ComboBoxItem>IT</ComboBoxItem>
        <ComboBoxItem>Marketing</ComboBoxItem>
        <ComboBoxItem>Finante</ComboBoxItem>
        <ComboBoxItem>HR</ComboBoxItem>
        <ComboBoxItem>Productie</ComboBoxItem>
        <ComboBoxItem>Logistica</ComboBoxItem>
      </ComboBox>

      <Label Content="Data angajării (de la):" Margin="0"/>
      <DatePicker x:Name="dpDataAngajarii"
SelectedDateChanged="DpDataAngajarii_SelectedDateChanged"
                  FontSize="14" Margin="0,0,0,10"/>
    </StackPanel>
  </Grid>
</Border>
```

Input Form with Specialized Controls

```
<!-- Input Form Section -->
```

```
<GroupBox Grid.Row="2" Header="Adăugare Angajat" Margin="0,10,0,0">
  <Grid Margin="5">
    <Grid.ColumnDefinitions>
      <ColumnDefinition Width="Auto"/>
      <ColumnDefinition Width="*"/>
      <ColumnDefinition Width="Auto"/>
      <ColumnDefinition Width="*"/>
    </Grid.ColumnDefinitions>
    <Grid.RowDefinitions>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
    </Grid.RowDefinitions>

    <!-- Row 0: Nume and Prenume -->
    <Label Grid.Row="0" Grid.Column="0" Content="Nume:"/>
    <TextBox Grid.Row="0" Grid.Column="1" x:Name="txtNume"
TextChanged="ValidateForm"/>
    <Label Grid.Row="0" Grid.Column="2" Content="Prenume:"/>
    <TextBox Grid.Row="0" Grid.Column="3" x:Name="txtPrenume"
TextChanged="ValidateForm"/>

    <!-- Row 1: Funcție and Departament -->
    <Label Grid.Row="1" Grid.Column="0" Content="Funcție:"/>
    <TextBox Grid.Row="1" Grid.Column="1" x:Name="txtFunctie"
TextChanged="ValidateForm"/>
    <Label Grid.Row="1" Grid.Column="2" Content="Departament:"/>
    <ComboBox Grid.Row="1" Grid.Column="3" x:Name="cmbDepartament" FontSize="14"
Padding="5">
```

```
        <ComboBoxItem>IT</ComboBoxItem>
        <ComboBoxItem>Marketing</ComboBoxItem>
        <ComboBoxItem>Finante</ComboBoxItem>
        <ComboBoxItem>HR</ComboBoxItem>
        <ComboBoxItem>Productie</ComboBoxItem>
        <ComboBoxItem>Logistica</ComboBoxItem>
    </ComboBox>

    <!-- Row 2: Tip Salariu RadioButtons -->
    <Label Grid.Row="2" Grid.Column="0" Content="Tip Salariu:"/>
    <StackPanel Grid.Row="2" Grid.Column="1" Orientation="Horizontal">
        <RadioButton x:Name="rbLunarInput" Content="Lunar" GroupName="TipSalariuInput"
Margin="5" Checked="RadioButtonInput_Checked"/>
        <RadioButton x:Name="rbOrarInput" Content="Orar" GroupName="TipSalariuInput"
Margin="5" Checked="RadioButtonInput_Checked"/>
    </StackPanel>

    <!-- Row 3: Salariu Brut / Tarif Oră -->
    <Label Grid.Row="3" Grid.Column="0" Content="Salariu Brut:"/>
    <TextBox Grid.Row="3" Grid.Column="1" x:Name="txtSalariuBrut"
TextChanged="ValidateForm"/>
    <Label Grid.Row="3" Grid.Column="2" Content="Tarif Oră:"/>
    <TextBox Grid.Row="3" Grid.Column="3" x:Name="txtTarifOra"
TextChanged="ValidateForm"/>

    <!-- Row 4: Ore Lucrate and Bonusuri -->
    <Label Grid.Row="4" Grid.Column="0" Content="Ore Lucrate:"/>
    <TextBox Grid.Row="4" Grid.Column="1" x:Name="txtOreLucrate"
TextChanged="ValidateForm"/>
    <Label Grid.Row="4" Grid.Column="2" Content="Bonusuri:"/>
    <TextBox Grid.Row="4" Grid.Column="3" x:Name="txtBonusuri"
TextChanged="ValidateForm"/>
```

```
    <!-- Row 5: Data Angajarii and Buttons -->
    <Label Grid.Row="5" Grid.Column="0" Content="Data Angajării:"/>
    <DatePicker Grid.Row="5" Grid.Column="1" x:Name="dpDataAngajarii" FontSize="14"
Padding="5" SelectedDate="{x:Null}"/>

    <StackPanel Grid.Row="5" Grid.Column="2" Grid.ColumnSpan="2"
Orientation="Horizontal" HorizontalAlignment="Right">
      <Button Content="Adaugă" x:Name="btnAdauga" Click="BtnAdauga_Click"
          Background="{StaticResource AccentBrush}" Foreground="White" Padding="15,8"
IsEnabled="False"/>
      <Button Content="Resetează" Click="BtnReset_Click"
          Background="{StaticResource PrimaryBrush}" Foreground="White"
Padding="15,8"/>
    </StackPanel>
  </Grid>
</GroupBox>
```

Code-Behind with Enhanced Filtering

```
private void AplicaFiltre()
{
  var rezultat = _angajatiOriginali.AsEnumerable();

  // Filter by salary type (RadioButton)
  if (rbLunar.IsChecked == true)
    rezultat = rezultat.Where(a => a.TipSalariu == TipSalariu.Lunar);
  else if (rbOrar.IsChecked == true)
    rezultat = rezultat.Where(a => a.TipSalariu == TipSalariu.Orar);

  // Filter by bonus (CheckBox)
  if (chkBonus.IsChecked == true)
```

```
    rezultat = rezultat.Where(a => a.Bonusuri > 0);


  // Filter by function (ListBox multiple selection)
  var functiiSelectate = lstFunctii.SelectedItems.Cast<ListBoxItem>().Select(item =>
item.Content.ToString()).ToList();
  if (functiiSelectate.Any())
  {
    rezultat = rezultat.Where(a => functiiSelectate.Contains(a.Functie));
  }


  // Filter by department (ComboBox)
  if (cmbDepartamentFiltru.SelectedItem is ComboBoxItem deptItem &&
deptItem.Content.ToString() != "Toate")
  {
    string dept = deptItem.Content.ToString();
    if (Enum.TryParse<Departament>(dept, out Departament departament))
    {
      rezultat = rezultat.Where(a => a.Departament == departament);
    }
  }


  // Filter by date (DatePicker)
  if (dpDataAngajarii.SelectedDate.HasValue)
  {
    DateTime dataFiltru = dpDataAngajarii.SelectedDate.Value;
    rezultat = rezultat.Where(a => a.DataAngajarii >= dataFiltru);
  }


  // Search by name (TextBox)
  if (!string.IsNullOrWhiteSpace(txtCautareNume.Text))
  {
```

```
        string searchTerm = txtCautareNume.Text.Trim();
        rezultat = rezultat.Where(a =>
            a.Nume.Contains(searchTerm, StringComparison.OrdinalIgnoreCase) ||
            a.Prenume.Contains(searchTerm, StringComparison.OrdinalIgnoreCase) ||
            a.NumeComplet.Contains(searchTerm, StringComparison.OrdinalIgnoreCase)
        );
    }

    dgAngajati.ItemsSource = rezultat.ToList();
}

private void LstFunctii_SelectionChanged(object sender, SelectionChangedEventArgs e)
{
    AplicaFiltre();
}

private void CmbDepartament_SelectionChanged(object sender,
SelectionChangedEventArgs e)
{
    AplicaFiltre();
}

private void DpDataAngajarii_SelectedDateChanged(object sender,
SelectionChangedEventArgs e)
{
    AplicaFiltre();
}

private void TxtCautare_TextChanged(object sender, TextChangedEventArgs e)
{
    AplicaFiltre();
```

}

Why I Used This Approach

ComboBox for Departments: The ComboBox displays a dropdown list of departments. This eliminates typos and ensures data consistency. The user can only select from predefined values.

DatePicker for Dates: The DatePicker provides a graphical calendar interface for selecting dates. This is much more user-friendly than asking users to type a date in a specific format. It also prevents invalid date entries.

ListBox for Multiple Selection: Setting `SelectionMode="Multiple"` allows users to select multiple job functions by holding Ctrl and clicking. This is useful for filtering by multiple criteria simultaneously.

Dynamic Filter Updates: Each filter control triggers a refresh of the DataGrid, providing immediate feedback to the user.

Combined Filters: All filters work together – the user can combine salary type, bonus, function, department, date, and name search simultaneously.

x:Null for DatePicker: Setting `SelectedDate="{x:Null}"` initializes the DatePicker with no date selected.

Enum Parsing: When the department ComboBox selection changes, the selected string is parsed to the corresponding enum value using `Enum.TryParse`.

Testing & Results

I tested the enhanced filtering:

1. **ComboBox**: Selecting "IT" from the department filter showed only IT employees
2. **DatePicker**: Selecting a date showed only employees hired on or after that date
3. **ListBox**: Selecting multiple functions showed employees who matched any of the selected functions
4. **Combined Filters**: Applying multiple filters at once worked correctly
5. **Input Form**: The ComboBox and DatePicker in the input form worked properly for adding new employees

All controls provided a smooth user experience with immediate feedback.

Reflection

WPF's built-in controls save a huge amount of development time. The ComboBox with enum binding is particularly elegant – I don't need to manually populate items or handle conversion. The DatePicker would have been complex to implement from scratch. The ListBox with multiple selection is a powerful filtering tool that would require significant code in other frameworks. These controls demonstrate WPF's strength in data-driven applications. The ability to combine multiple filters using LINQ's chaining capabilities makes the filtering logic both powerful and maintainable.